Dynamic Planet C

UT Invitational 2021

Time limit: 50 minutes

Team name: _____ (#_____)

Team members:

_____

_____

Directions

- *Do NOT open the packet until directed to do so by the tournament proctor(s).*
- Each team may bring a 2" or smaller three-ring binder and two stand-alone calculators of any type.
- You will have **50 minutes** to complete this test.
    - Some questions have "select ALL that apply." In **Section 1**, the point value (not given) for these questions is equal to the number of correct answers. In **Section 2**, the given point value is NOT necessarily equal to the number of correct answers. *One or more answers may apply.*
    - For these questions, correct answers earn 1 pt, incorrectly marked answers deduct 1 pt, and unmarked correct answers do not earn nor deduct points.
    - The minimum score per multiple-select question is zero.
- Please write the letter(s) of the answer(s) to each multiple-choice or multiple-select question IN CAPITAL LETTERS. No credit will be given for illegible or unclear answers.
- Tiebreakers are marked in the test in the order they will be broken (e.g., "TB1" is the first tiebreaker, "TB2" the second, etc.). Questions 60, 52, 28, 5, 16, 17, 34, 42, 67, 66

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Section 1	_____ / 45	
Section 2	_____ / 55	
TOTAL	_____ / 100	RANK _____

Section 1 [45 pts]

Reminder: Some questions with multiple correct answers are worth more than 1 pt each.

1. Near the headwaters of a stream in steep terrain, the size and shape of the sediment is most likely
    a. relatively small and angular
    b. relatively small and round
    c. relatively large and angular
    d. relatively large and round

2. According to the Strahler stream order system, the merging of two third-order streams results in a stream that is of what order?
    a. Second
    b. Third
    c. Fourth
    d. Fifth
    e. Sixth

3. If a region becomes afforested (has more vegetation), which choice best describes the changes to local runoff and transpiration?

	Runoff	Transpiration
a.	Increase	Increase
b.	Increase	No change
c.	Increase	Decrease
d.	Decrease	Increase
e.	Decrease	No change
f.	Decrease	Decrease

4. A river that can carry and ___ large particles, by definition, has a relatively high ___.
    a. erode, capacity
    b. erode, competence
    c. deposit, capacity
    d. deposit, competence

5. **[TB4]** Data collected from two different segments of the same stream are shown below.

Segment	Length of stream segment (km)	Length of straight-line distance (km)
A	2.5	1.5
B	3.5	2.5

    I. Segment A is more sinuous than Segment B
    II. Segment B is likely to be located downstream of Segment A

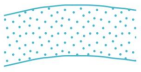
    Which of the statements above is/are true, if any?
    a. I, only
    b. II, only
    c. I and II
    d. Neither

6. Which of these is/are true about Earth's water and hydrologic cycle?
    I. In the oceans, overall precipitation balances overall evaporation
    II. Most freshwater is contained in glaciers and ice
    III. Energy from the Sun drives the hydrologic cycle

    a. I, only
    b. II, only
    c. III, only
    d. I and III, only
    e. II and III, only
    f. I, II, and III

Refer to **Image A** for questions 7 and 8.

7. What most likely shaped the landscape in the image?
    a. Deposition caused by low pH groundwater
    b. Deposition caused by high pH groundwater
    c. Deposition caused by hypersaline groundwater
    d. Erosion caused by low pH groundwater
    e. Erosion caused by high pH groundwater
    f. Erosion caused by hypersaline groundwater

8. Which topographic map symbol is most closely associated with this type of landscape?
    a. 
    b. 
    c. 
    d.

9. Which conditions below most favor valley deepening?
    a. A steep gradient with a high base level
    b. A steep gradient with a low base level
    c. A shallow gradient with a high base level
    d. A shallow gradient with a low base level

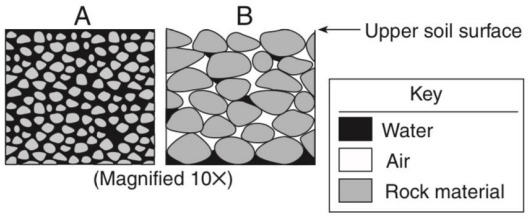
10. Which of the following correctly ranks delta beds from coarsest sediment to finest sediment?
    a. Topset, foreset, bottomset
    b. Topset, bottomset, forest
    c. Foreset, bottomset, topset
    d. Foreset, topset, bottomset
    e. Bottomset, foreset, topset
    f. Bottomset, topset, foreset

11. Atmospheric water content is about what percent of oceanic water content?
    a. Less than 1%
    b. 1%
    c. 5%
    d. 10%
    e. 20%

12. In the Midwestern US, what is the most likely consequence of excessive agricultural demand for groundwater?
    a. Freshwater acidification
    b. Saltwater intrusion
    c. Land subsidence
    d. Channel switching
    e. Eutrophication of lakes

13. Which of the following would provide the best aquifer?
    a. Mudstone
    b. Fractured granite
    c. Gravel mixture
    d. Unweathered clay

14. The diagram below shows two samples of soil 10 minutes after the end of a rainstorm.



What can be concluded about the soil samples?

I. The capillarity of A is greater than that of B
II. Surface runoff at B is greater than surface runoff at A

a. I, only
b. II, only
c. I and II
d. None of these

15. Select ALL of the following examples of nonpoint source pollution.
    a. Motor oil runoff from highways
    b. Pesticides from adjacent agricultural fields
    c. Wastewater from a factory pipe
    d. Sedimentation pollution along stream banks

16. **[TB5]** Select ALL of the following that increase in the downstream direction.
    a. Deposition minus erosion
    b. Channel size
    c. Channel gradient
    d. Channel roughness
    e. None of these

Use the following choices for 17-20. Each choice may be used once, more than once, or not at all.

a. Dissolved load
b. Suspended load
c. Bed load
d. None of these

17. **[TB6]** Which type(s) of load is/are affected by stream velocity? Select ALL that apply.

18. Which type of load is the main contributor to stream turbidity?

19. Pebbles and cobbles are most commonly part of which load?

20. Chemical pollutants, such as fertilizer, are commonly transported as part of which load?

21. Current meters can directly measure which of the following? Select ALL that apply.
    a. stage
    b. discharge
    c. gradient
    d. velocity
    e. none of these

22. Refer to **Image B**, a map of water table contours with a stream. The stream is best classified as a(n)
    a. gaining stream
    b. losing stream
    c. ephemeral stream
    d. disconnected stream

Refer to **Image C** for questions 23 and 24.

23. Which segment of the stream, if any, has a higher velocity?
    a. The purple segment
    b. The green segment
    c. Neither; both have the same velocity
    d. Cannot be determined from the map

24. Given that the top of the map is north, towards what direction is the groundwater most likely flowing at point P? Assume the stream is fed by groundwater.
    a. east
    b. southeast
    c. south
    d. southwest
    e. west

25. Which of these best describes the formation of kettle lakes?
    a. precipitation and hydrothermal fluid collect within a caldera
    b. melted glacial water fills an alpine depression formed by glacial erosion
    c. a segment from a meandering river cuts off, forming a crescent-shaped lake
    d. an ice block melts in an outwash plain and fills a depression

26. Select ALL of the following that apply to oligotrophic lakes.
    a. High nutrient concentrations
    b. High dissolved oxygen concentrations
    c. Generally low in turbidity
    d. High aquatic plant biodiversity

27. When lake waters warm in the summer, dissolved carbon dioxide levels tend to ___ and dissolved oxygen levels tend to ___.
    a. increase; increase
    b. increase; decrease
    c. decrease; increase
    d. decrease; decrease

28. **[TB3]** Refer to **Image D**. Select ALL statements that apply to this image.
    a. This channel type is more common than straight channels.
    b. Oxbow lakes are associated most closely with this channel type.
    c. This channel type can form due to variable discharge.
    d. This channel type is associated with gentle gradients.
    e. This channel type is associated with a high sediment load.

29. Select ALL of the following impacts of dam construction on a stream.
    a. Increased downstream water temperatures
    b. Increased downstream erosion
    c. Increased upstream erosion
    d. Increased upstream pollutant accumulation

30. The main source of water for bogs is ___, whereas the main source of water for marshes is ___.
    a. precipitation, groundwater
    b. precipitation, surface water
    c. groundwater, precipitation
    d. groundwater, surface water
    e. surface water, precipitation
    f. surface water, groundwater

31. Which of the following, if any, is/are disadvantages to injecting polluted water into a deep well?

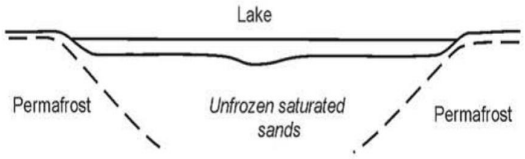
I. The polluted water will have a long residence time at depth
II. The polluted water can eventually contaminate surface water sources

    a. I, only
    b. II, only
    c. I and II
    d. Neither

32. A shallow-water wave on the surface of a lake depends on which of the following?
    a. Depth only
    b. Wavelength only
    c. Both depth and wavelength
    d. Neither depth nor wavelength

33. Select ALL statements that are true about knickpoints.
    a. Tectonic uplift can result in knickpoint formation
    b. Knickpoints can be created as a result of glacial activity
    c. Knickpoints are key features of graded streams
    d. Knickpoints are features contained within V-shaped valleys

34. **[TB7]** Consider the following cross-section of a high-latitude landscape.



Select ALL statements that are true related to the cross-section above.
    a. The diagram represents the initial stage of open-system pingo formation
    b. The pingo that will form will be a result of hydraulic pressure
    c. The cross-section indicates an area with abundant groundwater supply
    d. The permafrost likely exists as a continuous layer below the sands

35. Refer to **Image E**. Select ALL that apply to this image.
    a. The feature was likely formed by a waterfall
    b. Flash flooding risk at this feature is relatively high
    c. The feature represents a low-energy depositional environment
    d. The feature was likely the site of a former playa lake
    e. The feature was likely the site of a former river delta
    f. The sediment at A is coarser than the sediment at B

Section 2 [55 pts]

Questions 36 to 40: Refer to the drainage patterns shown in **Image F**.

36. Consider the pattern shown in Region 1.
    a. (1 pt) Identify the drainage pattern shown.

    b. (1 pt) Which shaded area, if any, has more resistant bedrock?
        A. Blue
        B. Green
        C. Each have bedrock of equal resistance
        D. There is not enough information; either could be more resistant

    c. (1 pt) Which statement is most likely true about groundwater flow in Region 1?
        A. Groundwater flows from the blue to the green area
        B. Groundwater flows from the green to the blue area
        C. Groundwater flows in both directions
        D. There is not enough information to infer the groundwater flow direction

(1 pt each) On the answer sheet, check off the region(s), if any, that apply to the following statements. One, multiple, or no regions at all may apply for each statement.

37. Most common of all drainage patterns

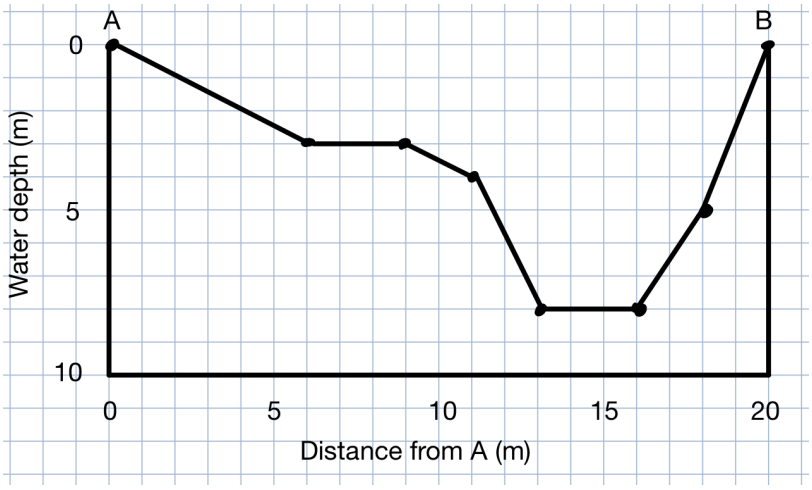
38. Convergence of streams into a central depression

39. Drainage pattern forms on a steeply sloping surface

40. Example of an antecedent drainage system

Questions 41 to 44

Refer to the stream profile below.



41. (1 pt) Which point on the stream profile has the highest stream velocity?
    A. distance from A: 8 m, depth: 1 m
    B. distance from A: 14 m, depth: 2 m
    C. distance from A: 15 m, depth: 6 m
    D. distance from A: 18 m, depth: 3 m

42. **[TB8]** (3 pts) Assuming that the average velocity of the stream is 5 m/s, calculate the flow rate of the stream. Show your work.

43. (1 pt) Is the point bar located at the left side or the right side of the stream profile? Is the sediment at a point bar likely to be well sorted or poorly sorted?

44. (2 pts) Suppose the base level of the stream dropped. Would the stream discharge increase, decrease, or remain unchanged? Briefly justify your answer.

Questions 45 to 51

Refer to **Image G**, a topographic map for questions 45-47. The 50-foot contour line is labeled. The contour interval is 10 feet.

45. (2 pts) Indicate whether point A is located upstream or downstream of point C. Briefly justify your answer.

46. (1 pt) Suppose the water table depth at point B is 10 feet below the surface. State one possible whole number value of the water table elevation at point B.

47. (1 pt) On some months, an ephemeral lake can form at point B. Briefly describe the meaning of any clues on the topographic map that may lead to this inference.

The following data table shows the precipitation (P) and potential evapotranspiration (E_p) (in mm) for each month in the region.

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
P	73	55	34	23	17	23	31	38	43	55	63	77
E_p	29	33	56	85	135	147	156	151	122	88	62	41

48. (2 pts) Based on the data above, briefly describe what calculations are needed to determine the climate type (arid, semiarid, subhumid, or humid) of the region. (You do *not* need to perform the calculations)

49. (1 pt) During which month does the groundwater begin to recharge again?

50. (1 pt) Assuming precipitation and evapotranspiration are the only inputs and outputs to the ephemeral lake, during which month is the lake level likely to be the highest?

51. (1 pt) Which of the following best describes the relationship between the actual evaporation rate (E) and the potential evapotranspiration (E_p)?
    a. $E > E_p$
    b. $E = E_p$
    c. $E < E_p$
    d. $E \geq E_p$
    e. $E \leq E_p$

Questions 52 to 56

Refer to **Image H**, a cross-sectional profile of a typical floodplain. Some features are labeled; assume all features are formed naturally.

52. **[TB2]** (3 pts) Identify the features x, y, and z. Assume feature z is NOT part of a lake. (Hint: Feature x is a landform corresponding to the purple surface - NOT "floodplain," y is NOT "riverbank," and there is a specific name for feature z).

53. (1 pt) Write the letter(s) of the features (x, y, z) which formed as the direct result of flooding.

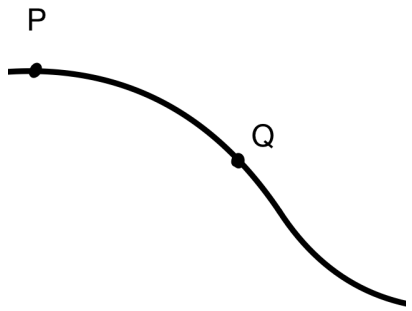
54. Suppose zone x is designated the 200-year floodplain.
    a. (1 pt) What is the probability of the occurrence of a flood that affects zone x each year? Give your answer as a percent.

    b. (1 pt, no partial credit) Select ALL of the following true statements.
        A. Multiple 200-year floods can recur within the period of the recurrence interval (i.e., within a 200-year period).
        B. Changes in land use can increase the probability of a 200-year flood.
        C. The probability of a 200-year flood occurring in a 200-year period is 100%.

55. (1 pt) Channelization is one approach to controlling floods. Channels that are built to minimize flooding risk are designed to ____ stream velocity and ____ cross-sectional area.
    a. reduce; reduce
    b. reduce; maximize
    c. maximize; reduce
    d. maximize; maximize

56. Storm hydrographs, graphs of discharge versus time, can be used to assess flooding risk for different areas.
    a. (1 pt) Suppose a heavy rainstorm begins at 7am and ends at 10am. The peak of the hydrograph for an urbanized area occurs at 11am, and the discharge returns to pre-storm levels by 5pm. What is the lag time?
        A. 1 hour
        B. 3 hours
        C. 4 hours
        D. 6 hours
        E. Not enough information

    b. (2 pts) Consider a hill of uniform material with two points, P and Q, as shown below. On the axes provided on the answer sheet, sketch and label clearly two storm hydrographs for point P and point Q. On the *x* axis, label the time of peak discharge for each curve. Assume that both locations receive the same amount of rain over the same time.



Questions 57 to 62

Refer to **Image I**, a cross section with rock units shown. The blue surface denotes the water table, and the green dashed line denotes the potentiometric surface. Wells A, B, C, and D are labeled, and w, x, y, and z refer to geologic contacts along a steep slope.

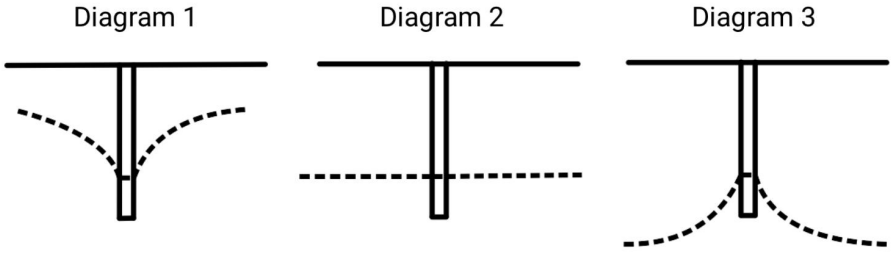
57. (1 pt) What is responsible for the upward movement of groundwater in the wells? (A word or phrase is enough)

58. (2 pts) Identify the letter(s) of the artesian wells. Briefly justify your answer.

59. (1 pt) Identify the specific type of aquifer that Well C is located in. (Hint: This type of aquifer is unique to Well C.)

60. **[TB1]** (3 pts) Assume that all of the wells do NOT have artificial pumps. Predict where the water level in each well B, C, and D will rise to by putting a check for each row in the table provided in the answer sheet.

61. (1 pt) If groundwater is always being pumped upward from a well at a nonzero rate, which diagram(s) best show the position of the water table (indicated by the dashed lines)?



a. Diagram 1
b. Diagram 2
c. Diagram 3
d. Diagram 1 or 3, depending on the pumping rate
e. Diagram 1 or 3, depending on the hydraulic conductivity of the nearby rock

62. (2 pts) Of the geologic contacts, identify the letter(s) in which (a) spring(s) is/are likely to flow.

<u>Questions 63 to 66</u>
Lakes have several physical and chemical properties.

63. Consider a mid-latitude dimictic lake (a lake that mixes twice a year). Let the epilimnion temperature and density of the lake be T_E and ρ_E, and the hypolimnion temperature and density of the lake be T_H and ρ_H, respectively.
    a. (1 pt) During the winter, which of these is true about the temperature?
        A. $T_E > T_H$
        B. $T_E < T_H$
        C. $T_E = T_H$

    b. (1 pt) During the winter, which of these is true about the density?
        A. $\rho_E > \rho_H$
        B. $\rho_E < \rho_H$
        C. $\rho_E = \rho_H$

64. (1 pt) In a dimictic lake, during which season(s) is dissolved oxygen content greatest in the hypolimnion layer?

65. (1 pt) The total phosphorus content in a lake is ____ related to the chlorophyll *a* content and ____ related to the total light levels in a lake.
    a. directly; directly
    b. directly; inversely
    c. inversely; directly
    d. inversely; inversely

66. **[TB10]** Sometimes, waves have been observed to oscillate back and forth in lakes.
    a. (1 pt) What is the name given to this type of wave?

    b. (2 pts) Select ALL of the true statements regarding this phenomenon.
        A. These waves are primarily driven by pressure differences
        B. These waves are primarily driven by tidal forces
        C. These waves are primarily wind-driven
        D. These waves typically last for a few minutes
        E. These waves can cause lakeshore flooding

67. **[TB9]** (2 pts) Positive feedback loops amplify or enhance changes, whereas negative feedback loops reduce or dampen the initial change. Classify the feedback loop between soil moisture content and evapotranspiration as a positive or negative feedback loop, and briefly justify your answer.

68. Past glaciation events can have a significant impact on the hydrology of an area.
    a. (1 pt) Is glacial sediment well or poorly sorted? How does the sorting affect the hydraulic conductivity of the sediment?

    Refer to **Images J** and **K** (adapted from Fergus et al., 2017). The solid black line represents the estimated boundary of the Wisconsin glacial period. The legend for stream connectivity (Image K) has been omitted.
    b. (1 pt) Based on the figures, what is the relationship between bog and marsh density and whether the area has been affected by glaciation?

    c. (1 pt) Does the dark purple shading in Image K indicate low-order dominated or high-order dominated stream connectivity?

    d. (1 pt) Do streams south of the glacial boundary have more or less tributaries than streams north of the glacial boundary? Are watersheds south of the glacial boundary more well drained or less well drained than those north of the boundary?